

Smart Decision Making Using Data Science Technology with COVID-19 Data *

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Abstract—The rapid spread of the novel coronavirus COVID-19 since December 2019 has posed unprecedented challenges to healthcare systems and decision-makers in public health. The responses to the virus differed from country to country, which led to variable outcomes throughout. Understanding how the answer to the virus and resource availability affected the outcomes is important. In this paper, we present our work on using various data science technologies including correlation analysis, regression analysis, and exploratory data analysis to understand the relationship among different variables and factors. We also present a set of data visualization tools that consist of two types of geospatial visualization functions, standard bar chart, stacked bar chart, and line chart. These tools help users to better understand the data and to make smart decisions in policies related to response, prevention, and intervention of pandemic episodes.

Index Terms—Data Analytics, Data Visualization, Healthcare Decision Support

I. INTRODUCTION

Computer technologies have been used in healthcare and medical domains with many functions including information storage and retrieval, knowledge management, image processing, and decision support. The rapid spread of the novel coronavirus COVID-19 since December 2019 has posed unprecedented challenges to healthcare systems and decision-makers in public health. The responses to the virus differed from country to country, which led to variable outcomes throughout. It is important to understand how the answer to the virus alongside resource availability affected the outcomes.

Understanding the pandemic will help take the right measures at the right time. For example, travel advisories at the beginning of an outbreak may help slow the spread of contagious diseases. Taking these measures before the outbreak of the disease will help maintain the health and economy of the country. With the continuous spread of the epidemic, many types of information and data related to the epidemic are continually updated on the Internet. There is a huge amount of information, including global epidemic data, COVID-19-related research statistics, medical equipment details, transportation information, etc. The integration and classification of these forms of information are important for epidemic prevention and control.

Data science plays a pivotal role in understanding the COVID-19 pandemic data. It is used to analyze the data both

structured and unstructured, potential insights could be drawn using this. Data science provides support in dealing with the pandemic and helps in decision-making in many areas including risk assessment, patient prioritization, screening, diagnosis, contact tracing, automated patient care, vaccine discovery, and economic interventions. However, these applications demand technical skills in computer and data sciences, which are not necessarily available to healthcare professionals and public health staff.

The website Our World in Data [1] provides data on the coronavirus pandemic and a number of data visualization tools to help users to visualize the data on certain selected topics such as the number of cases, deaths, vaccinations, testing cases, policy responses, and excess mortality. These are very useful tools for users to directly visualize large and daily updated datasets. However, there are also limitations to these tools. First, these tools are mostly for data visualization and no data analysis tools have been provided such as to understand the correlation between different variables, such as positive rate, the number of hospital patients per million, and the total vaccination per hundred. Secondly, only a limited number of visualization templates are provided - users may adjust some parameters but are not able to create their own templates.

In this paper, we present our work in developing interactive web applications with various data analysis and visualization functions for covid dataset. Four main functionalities are provided. The first is the correlation analysis tool to help understand how different policies and actions were instrumental in affecting metrics like hospitalizations, vaccinations, etc. The second function is linear regression analysis which allows users to analyze the relationship between variables. The third function is exploratory analysis to provide a better understanding of dataset variables and the relationships among them. The last function is data visualization including geospatial visualization which shows the relationship between data and its physical location to generate insight.

The rest of the paper is organized as the following. We first present the system architecture and workflow in Section 2. We then discuss the process of acquiring data and preparing data in Section 3. Section 4 focuses on the data analysis functions and Section 5 describes the data visualization functions. In the end, section 6 concludes and discusses future work.

II. SYSTEM ARCHITECTURE AND WORKFLOW

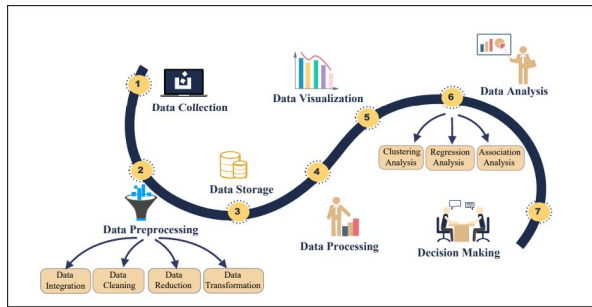


Fig. 1. Process from Data to Decision

The process to make a data-driven decision involves several steps as depicted in Figure 1. After data are collected from different sources, they need to be cleaned, integrated, reduced, and transformed according to specific standards and design criteria, this step is referred to as data pre-processing. In this work, we use the datasets from *Our World in Data* [1]. The data after pre-processing are stored in a specific format in a specific location. The stored data then are processed to generate visualized results and analyzed according to the request of users. The visualization and analysis results are then used to help users to make decisions.

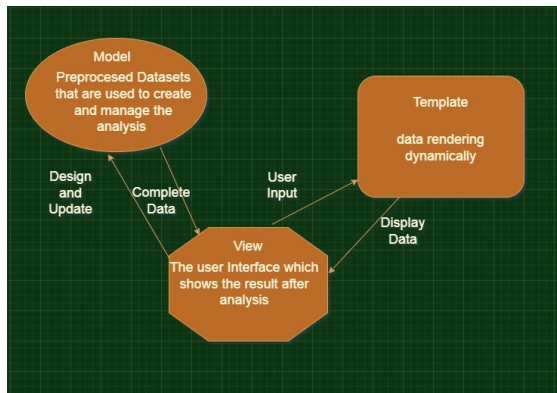


Fig. 2. System Architecture

The work presented in this paper is mainly focused on Step 5 Data Visualization and Step 6 Data Analysis. Step 4 Data Processing is necessary to achieve the purpose. The developed system adopts the MVT (Model-View-Template) architecture as shown in Figure 2. MVT is a software design pattern for developing a web application, which contains the following three parts:

- **Model:** served as the data interface. It is responsible for maintaining data. It is the logical data structure behind the entire application and is represented by a dataset.
- **View:** served as the user interface. It describes what the user sees in the web browser. It is represented by *HTML/CSS/Javascript* and *Jinja*[2] files.

- **Template:** consists of static parts of the desired *HTML* output as well as some special syntax describing how dynamic content will be inserted.

III. DATA ACQUISITION AND PREPARATION

The data used in this work was acquired from our world data [1]. After retrieving all the records, the next step is to preprocess the data. This includes data cleansing, data consolidation, and data reduction. A dataset named *owid-covid-data* [3] is created and made available online for public access.

The first step in data preparation is to consolidate multiple datasets to construct one complete dataset for the purpose of this study. In this process, data records from various datasets need to be matched and joined correctly. Additionally, irrelevant data and duplicates need to be removed as well. For example, the dataset *excess-mortality-p-scores-projected-baseline* [4] contains the columns named as *entity*, *country code*, *day* and *P-proj-all-ages* (P-score) as shown in Figure 3. The P-score is the percentage difference between the reported number of weekly or monthly deaths from 2020 to 2022 and the projected number of deaths for the same period based on previous years [4]. It can have different values either positive or negative. The entity column is the name of the country. For example, in the second line in Figure 3, entity Albania has code as ALB which is the abbreviation of the country. The next column Day has a value of 2020-01-31. The last data column is the *p-proj-all-ages* with a value of -10.16, meaning the excess mortality number in this period is about 10% less than in previous years.

Also, the dataset *covid-19-testing-policy-policy-response* [6] contains the columns named *Entity*, *country Code*, *Day* and *testing_policy*. The data *testing_policy* contains a value between 0 and 3, describing four different testing policies [7]:

- 1) no testing policy.
- 2) testing only those people who have symptoms and also meet specific criteria.
- 3) testing only those people who have symptoms.
- 4) open public testing available to asymptomatic people.

Figure 4 is showing the initial part of the dataset *covid-19-testing-policy-policy response*. The entity is the name of the country which is in the shown cases "Afghanistan". Then the other attribute is *Code*, which is an abbreviation of the country. *Day* is the date at which the data is gathered. The last column *testing_policy* has a value of 0 meaning that Afghanistan does not have a testing policy from January 1st to January 12 in 2020.

Similarly, Figure 5 shows the first 14 lines of the other data set that we have worked with is *current-covid-patients-hospitalizations* [8]. The first attribute entity is the country name, and the value is Australia in shown cases. The next is code which again is an abbreviation and it is AUS. *Day* is the date on which it is recorded. The first line of data shows that the number of hospitalizations and intensive care (ICU)

1	Entity	Code	Day	p_proj_all_ages
2	Albania	ALB	2020-01-31	-10.16
3	Albania	ALB	2020-02-29	2.88
4	Albania	ALB	2020-03-31	1.31
5	Albania	ALB	2020-04-30	4.04
6	Albania	ALB	2020-05-31	7
7	Albania	ALB	2020-06-30	10.22
8	Albania	ALB	2020-07-31	23.69
9	Albania	ALB	2020-08-31	46.11
10	Albania	ALB	2020-09-30	30.55
11	Albania	ALB	2020-10-31	45.55
12	Albania	ALB	2020-11-30	120.15
13	Albania	ALB	2020-12-31	54.6
14	Albania	ALB	2021-01-31	23.56
15	Albania	ALB	2021-02-28	92.8

Fig. 3. Excess Mortality Dataset

1	Entity,Code,Day,testing_policy
2	Afghanistan,AFG,2020-01-01,0
3	Afghanistan,AFG,2020-01-02,0
4	Afghanistan,AFG,2020-01-03,0
5	Afghanistan,AFG,2020-01-04,0
6	Afghanistan,AFG,2020-01-05,0
7	Afghanistan,AFG,2020-01-06,0
8	Afghanistan,AFG,2020-01-07,0
9	Afghanistan,AFG,2020-01-08,0
10	Afghanistan,AFG,2020-01-09,0
11	Afghanistan,AFG,2020-01-10,0
12	Afghanistan,AFG,2020-01-11,0
13	Afghanistan,AFG,2020-01-12,0

Fig. 4. COVID-19 Testing Policies in 38 Countries

1	Entity,Code,Day,Daily hospital occupancy
2	Australia,AUS,2020-03-31,426
3	Australia,AUS,2020-04-01,441
4	Australia,AUS,2020-04-02,415
5	Australia,AUS,2020-04-03,457
6	Australia,AUS,2020-04-04,490
7	Australia,AUS,2020-04-05,457
8	Australia,AUS,2020-04-06,305
9	Australia,AUS,2020-04-07,263
10	Australia,AUS,2020-04-08,262
11	Australia,AUS,2020-04-09,247
12	Australia,AUS,2020-04-10,240
13	Australia,AUS,2020-04-11,239
14	Australia,AUS,2020-04-12,227
15	Australia,AUS,2020-04-13,230

Fig. 5. COVID-19 hospitalizations

admissions due to COVID-19 on 2020-03-31 in Australia is 426.

After obtaining these datasets, the next step is to fix the structural error and deal with missing data. There are different data columns and data formats in these datasets, so we need first change them to the same format. In some datasets, there is some data missing or not available. Such data would cause imputed results if directly feed in the next step of the process, hence we need to replace them as 0 or nan value in order to get the correct result.

Furthermore, the next step is to filter out the data outliers and validate the data. In this step, we deleted the records that contain incorrect information. For example, in the dataset covid-19-testing-policy-policy-response [9], the data column is *Stringency Index* that measures the strictness of the policy. Some countries have loose policies or some counties have policies in place but geographically and population wise it is difficult to implement the policy. So we remove such records from the datasets. And the last part is to validate the data and use it in downstream processing.

The common columns of these individual datasets are *Entity*, country *Code*, and *Day*. Based on these common columns, multiple individual datasets are combined together to create a single dataset which is *owid-covid-data* [3]. Figure 6 shows the initial part of the final dataset which is obtained after the preprocessing of the data from individual datasets.

IV. DATA ANALYSIS FUNCTIONS

The Data Analysis Tools are implemented with Python [9] language and users may access these tools from a web interface [10]. Figure 7 shows this interface that provides three main analysis tools: correlation analysis, data exploratory and regression analysis. Next, each of these functions will be discussed in detail.

A. Correlation Analysis

Correlation analysis is to find out whether two variables covary; when one variable increases, whether, the other variable tends to increase or decrease. The correlation coefficient

Fig. 6. COVID-19 Final Dataset

Fig. 7. Data Analysis Web Interface

r is a measure of the strength of the correlation, which usually takes a value between -1.0 and 1.0 . -1.0 indicates a perfect negative correlation (as one variable increases the other variable decreases), 1.0 indicates a perfect positive correlation (as one variable increases the other variable also increases), and 0 indicates no correlation.

There are different ways to calculate the correlation coefficient. Currently, Spearman's Rank Correlation Coefficient [11] r_s is adopted in this project to measure the relationship between two ranked variables, which is calculated as:

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i is the difference between the two ranks of each

observation and n is the number of observations. In the future, it would be considered to include other correlation measures on the interface and allow the user to select the one that best fits the specific problem context.

Users may use this interface to choose two variables and then find out the correlation between these two variables. Table I shows four example outputs of this tool.

B. Regression Analysis

Linear Regression [12] is a statistical technique used to analyze relationships between variables. It has one or more dependent variables y and one or more independent variables X . It aims to construct a linear model that conforms to a straight-line equation: $y = MX + c$, where M is the coefficient vector and c is the in Each independent variable has

TABLE I
EXAMPLES OF CORRELATION ANALYSIS

Variable A	Variable B	r_s	Interpretation
positive rate	total vaccination per hundred	0.147	very weak positive correlation
hospital patients per million	total vaccination per hundred	-0.234	very weak negative correlation
weekly icu admission	weekly hospital admission	0.902	strong positive correlation
stringency index	excess mortality cumulative per million	-0.036	no correlation

OLS Regression Results						
Dep. Variable:	excess_mortality_cumulative_per_million	R-squared:	0.851			
Model:	OLS	Adj. R-squared:	0.851			
Method:	Least Squares	F-statistic:	6799.			
Date:	Fri, 06 May 2022	Prob (F-statistic):	0.00			
Time:	03:12:02	Log-Likelihood:	-2.1624e+05			
No. Observations:	35666	AIC:	4.325e+05			
Df Residuals:	35635	BIC:	4.328e+05			
Df Model:	30					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-3.4832	1.371	-2.541	0.011	-6.170	-0.797
total_cases_per_million	-1.394e-05	2.05e-05	-0.680	0.496	-5.41e-05	2.62e-05
new_cases_per_million	-0.0155	0.002	-8.794	0.000	-0.019	-0.012
new_cases_smoothed_per_million	0.0266	0.003	10.232	0.000	0.022	0.032
total_deaths_per_million	0.0137	0.002	8.481	0.000	0.011	0.017
new_deaths_per_million	0.7371	0.244	3.022	0.003	0.259	1.215
new_deaths_smoothed_per_million	-3.1569	0.368	-8.576	0.000	-3.878	-2.435
icu_patients_per_million	0.0189	0.117	0.161	0.872	-0.211	0.249
hosp_patients_per_million	0.0671	0.022	3.085	0.002	0.024	0.110
weekly_icu_admissions	-0.2379	0.009	-26.511	0.000	-0.256	-0.220
weekly_icu_admissions_per_million	12.1029	0.366	33.100	0.000	11.386	12.820
weekly_hosp_admissions_per_million	-0.2614	0.023	-11.613	0.000	-0.306	-0.217
total_tests	1.107e-09	1.1e-08	0.100	0.920	-2.05e-08	2.27e-08
total_tests_per_thousand	0.0092	0.001	16.508	0.000	0.008	0.010
positive_rate	15.0070	8.020	1.871	0.061	-0.712	30.726
tests_per_case	-0.0039	0.002	-2.053	0.040	-0.008	-0.000
total_boosters_per_hundred	0.1068	0.083	1.294	0.196	-0.055	0.269
new_vaccinations_smoothed_per_million	0.0005	0.000	1.631	0.103	-0.000	0.001
new_people_vaccinated_smoothed_per_hundred	-1.4510	5.763	-0.252	0.801	-12.746	9.844
population	-4.748e-11	4.65e-10	-0.102	0.919	-9.59e-10	8.64e-10
population_density	0.0049	0.001	7.080	0.000	0.004	0.006
gdp_per_capita	-0.0006	5.27e-05	-10.987	0.000	-0.001	-0.000
extreme_poverty	0.0557	0.041	1.351	0.177	-0.025	0.137
diabetes_prevalence	-0.1386	0.130	-1.063	0.288	-0.394	0.117
female_smokers	0.0932	0.130	0.715	0.475	-0.162	0.349
male_smokers	0.0174	0.040	0.433	0.665	-0.061	0.096
handwashing_facilities	0.0206	0.022	0.948	0.343	-0.022	0.063
hospital_beds_per_thousand	1.7095	0.289	5.915	0.000	1.143	2.276
excess_mortality_cumulative_absolute	0.0010	2.06e-05	48.272	0.000	0.001	0.001
excess_mortality_cumulative	113.1673	0.384	294.547	0.000	112.414	113.920
excess_mortality	-3.9640	0.155	-25.642	0.000	-4.267	-3.661
Omnibus:	16608.282	Durbin-Watson:	2.073			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	47264893.211			
Skew:	0.465	Prob(JB):	0.00			
Kurtosis:	181.337	Cond. No.	1.82e+10			

Fig. 8. Regression Analysis Results

a coefficient attached, and its significance can be interpreted by its corresponding p-value.

Figure 8 shows the regression analysis result with *excess_mortality_cumulative_per_million* as the dependent variable y . Independent variables X are chosen based on the Variance Inflation Factor (VIF) [13]. $VIF < 10$ is the threshold for selecting variables for the final analysis. After running the regression analysis, we see a decent R^2 (R_squared) value of 0.851, which shows a relatively strong relationship. However, we also noticed that the significant relationship is contributed by *excess_mortality_cumulative*, which is naturally correlated to the dependent variable in this analysis. Therefore,

the analysis with the selected X variables does not help much in discovering the interesting correlation relationship. A better use of this regression analysis would need the user to handpick X variables that are considered to be possibly strongly connected with y and then use this tool to validate this hypothesis.

C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is to analyze data sets and summarize their main characteristics, usually employing data visualization methods. EDA could be used to find what data can reveal and provide a better understanding of dataset

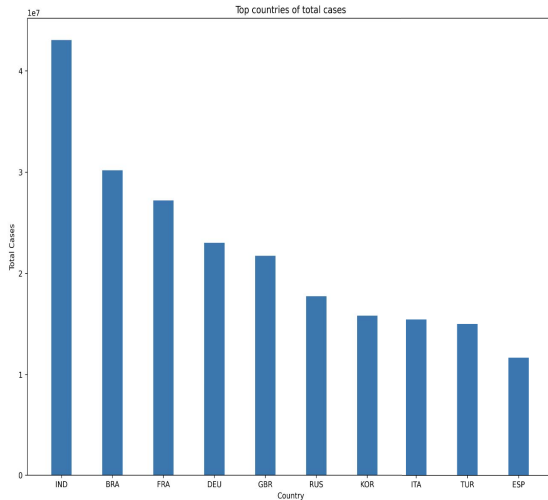


Fig. 9. Top 10 Countries by Total Cases: IND, BRA, FRA, DEU, GBR, RUS, KOR, ITA, TUR, and ESP

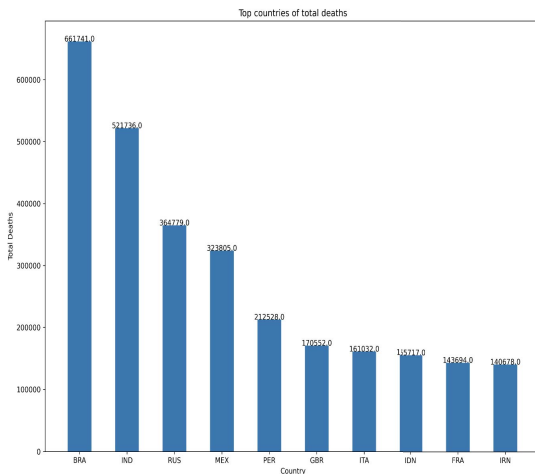


Fig. 10. Top 10 Countries by Total Deaths: BRA, IND, RUS, MEX, PER, GBR, ITA, IDN, FRA and IRN

variables and the relationships among them. It can also help users to choose appropriate statistical techniques for data analysis.

For example, Figure 9 shows the top 10 countries by total number of COVID-19 cases, and Figure 10 shows the top 10 countries by total number deaths related to COVID-19. By a quick comparison, it is easy to notice that #1 and #2 countries switch their places in these two graphs. To be more specific, India has more COVID cases but fewer deaths compared to Brazil. This finding may lead a user to further investigate other data on these two countries and to find what may be the causes for this.

Another EDA tool is Time Series Analysis, which is a

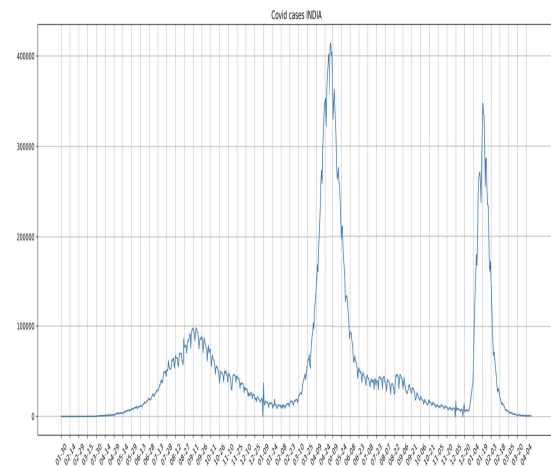


Fig. 11. Covid Cases in India

sequence of recorded data over regular time intervals. For example, Figure 11 shows the number of COVID cases in India over bi-weekly intervals. It is clear that there is one small spike and two very sharp high spikes. Further investigation of the relevant events around these three time periods may shine some light on the causes of these sudden increases in case numbers. Other analyses based on Time Series such as Moving Average (MA) and ARIMA (Auto-Regressive Integrated Moving Average) have also been implemented in this work. Space limitation prohibits more discussion here.

V. DATA VISUALIZATION FUNCTIONS

With the same dataset *owid-covid-data* [3], the following four visualization functions are provided through a web interface [14].

- Geospatial Visualization 1: visualize the global covid-19 data on a world map. As the user moves the cursor over the map, the related data of the current country is displayed on the screen. Figure 12 shows the number of confirmed cases, deaths per million people, and fully vaccinated percentage in the United States.
- Geospatial Visualization 2: color the map according to the value of the selected variable (total confirmed cases, total death cases, or vaccinations) with a particular time period (2020, 2021, or 2022). Figure 13 shows the total confirmed death cases in 2021 of every country, darker color represents greater values. Similarly, Figure 14 shows the vaccinations in 2021.
- Stacked Bar Chart Visualization: this Chart includes the visualization of vaccination rate & booster rate of all countries. This chart also contains other vaccination data sets such as the Top 30 countries' vaccination data, the Top 30 countries' booster data, etc. Figure 15 shows the top 30 countries according to the number of initial doses and boosters per 100 persons.

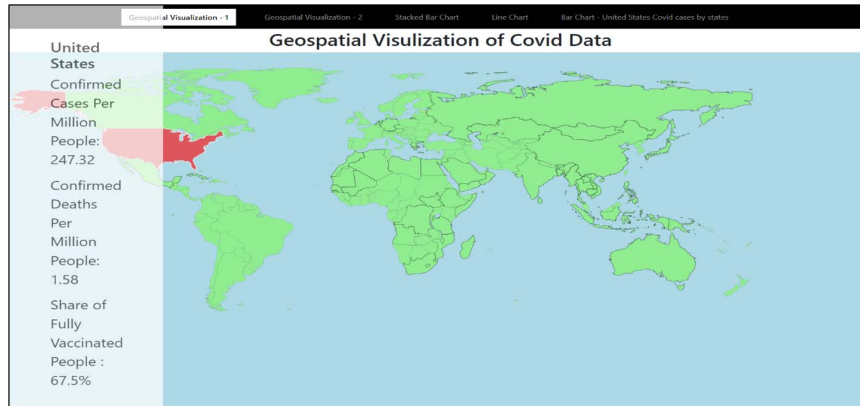


Fig. 12. Geospatial Visualization of Covid Data

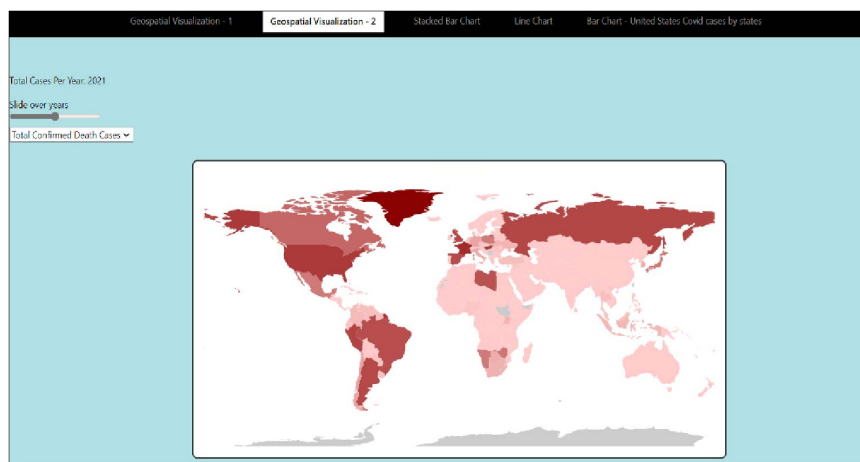


Fig. 13. Geospatial Visualization of Total Confirmed Death Cases in 2021



Fig. 14. Geospatial Visualization of Total Vaccinations in 2021

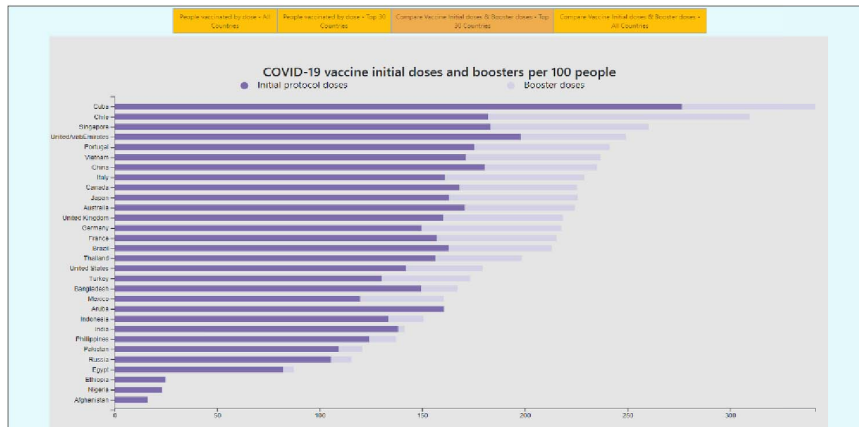


Fig. 15. Top 30 countries by Initial doses and Boosters per 100 Persons

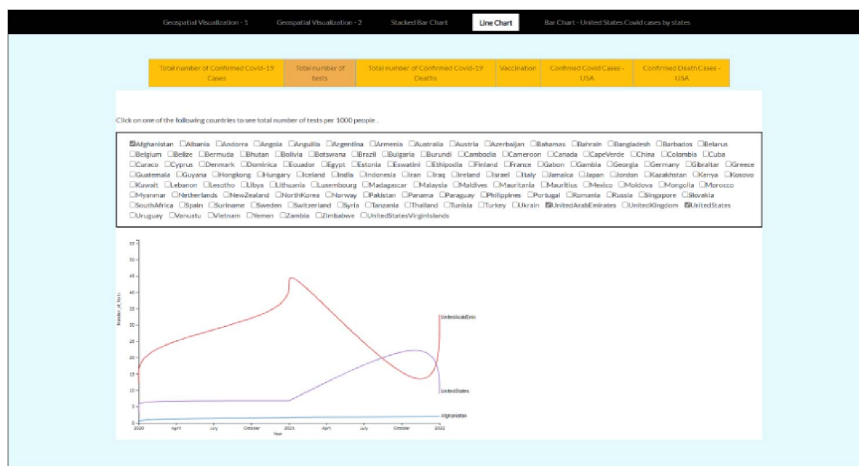


Fig. 16. Total Number of Tests per 1000 Persons for 3 Selected Countries

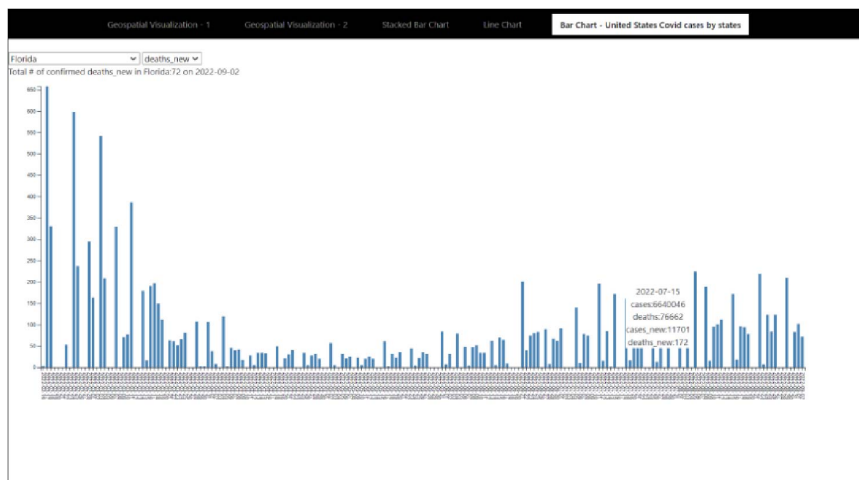


Fig. 17. Total Confirmed New Death Cases in Florida

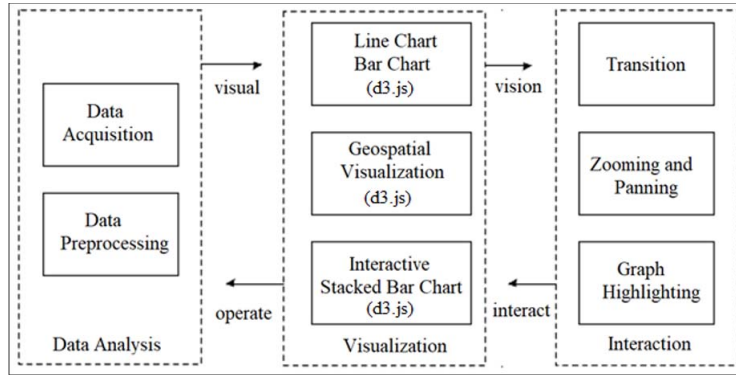


Fig. 18. General Visualization System Process flow

- Line Chart Visualization: the Line chart visualizations world data in different aspects such as numbers of test cases, confirmed cases, death cases, or vaccination rate. Line chart also visualizes united states data such as confirmed cases and death cases. Users may select which countries to be included in this chart. Figure 16 shows the total number of tests per 1000 persons for Afghanistan, the United Arab Emirates, and the United States.
- Bar chart Visualization: this visualization mainly focused on the covid-19 situation in the United States. It provides a bar chart representation of all states of the United States and this chart also consists of multiple variables such as confirmed cases, death cases, new cases, new deaths, etc. Figure 17 shows the total confirmed new death cases in the state of Florida.

Figure 18 depicts the visualization system process flow. We used HTML and CSS to implement the User Interface of the web application. D3.js [15] is a JavaScript library that handles documents based on data. D3 is a very powerful visualization tool that generates interactive data visualization in a web browser. D3 uses static as well as dynamic data in different formats like Arrays, JSON, Objects, CSV, XML.

VI. CONCLUSION AND FUTURE WORK

In this paper, we present our work on a set of web-based interactive data analysis and visualization tools for COVID-19 data. We discussed how the data is obtained, and pre-processed and how these tools are developed. This set of tools provides a broad range of functions including correlation analysis, regression analysis, and exploratory data analysis to understand the relationship among different variables and factors. Additionally, this tool set has a supply of standard bar charts, stacked bar charts, line charts, and two types of geospatial visualization functions. These data analysis and visualization tools help users to analyze and visualize data, better understand data, and discover the correlation relationship between variables, hence making smart decisions based on the results.

In the future, we plan to incorporate more interactive functions in the system, allowing users to select particular variables to analyze their relationships. More analysis tools may also be added to better serve the needs of the users. Incorporating dynamically updated data in the dataset would be another important task. Though the COVID-19 pandemic crisis has passed, the data-driven decision-making capability is still highly valuable in other similar scenarios. Once enabled with the capability of loading dynamically updated datasets, the interactive data analysis and visualization tools presented in this paper may be used in many other applications.

REFERENCES

- [1] Our World in Data, <https://ourworldindata.org/coronavirus>
- [2] Jinja Documentation, <https://jinja.palletsprojects.com/en/3.1.x/>
- [3] Dataset: ovid-covid-data, <https://github.com/arathi5056/datasets/blob/master/ovid-covid-data.csv>
- [4] Dataset: excess-mortality-p-scores-projected-baseline, <https://github.com/arathi5056/datasets/blob/master/excess-mortality-p-scores-projected-baseline.csv>
- [5] Excess mortality P-scores, <https://ourworldindata.org/excess-mortality-covid#excess-mortality-p-scores>
- [6] Dataset: covid-19-testing-policy-policy-response, https://github.com/arathi5056/datasets/blob/master/covid-19-testing-policy_policy%20response.csv
- [7] Testing policy, <https://ourworldindata.org/covid-testing-contact-tracing#testing-policy>
- [8] Coronavirus (COVID-19) Hospitalizations, <https://ourworldindata.org/covid-hospitalizations>
- [9] Python programming language, <https://www.python.org/>
- [10] Data Analysis Tools, <https://github.com/arathi5056/corelationproject>
- [11] Spearman Rank Correlation Coefficient, in The Concise Encyclopedia of Statistics. Springer, New York, NY. 2008. https://doi.org/10.1007/978-0-387-32833-1_379
- [12] Seltman, H. J., Experimental Design and Analysis, Chapter 9. Pittsburgh: Carnegie Mellon University. 2012.
- [13] Stephanie Glen. "Variance Inflation Factor", StatisticsHowTo.com: Elementary Statistics for the rest of us! <https://www.statisticshowto.com/variance-inflation-factor/>
- [14] Data Visualization Interface, <https://shivpatil254.github.io/coviddatavis.github.io/index.html>
- [15] D3 Data Driven Documents, <https://d3js.org/>